

Group name

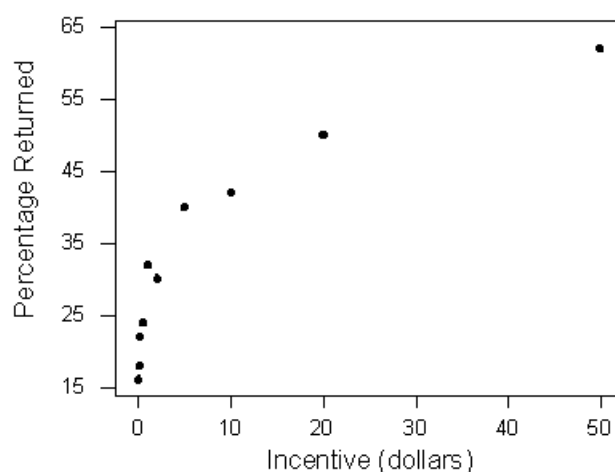
Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Reflect and discuss (A)

A study examined how the percentage of participants who completed a survey is related to the monetary incentive that researchers promised to participants. Consider the relationship between these two quantitative variables, displayed in the scatterplot below.



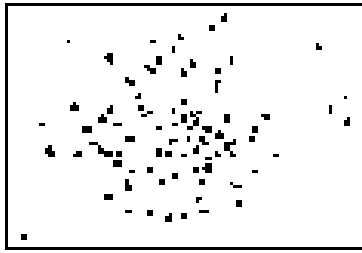
-  1. The _____ of this relationship is { positive, negative, neither positive nor negative }.
-  2. In the context of this example, this means that when researchers promised higher payments, the percentage of participants who _____ the survey { increased, remained the same, decreased }.
-  3. The _____ of the relationship is { linear, curvilinear, neither linear nor curvilinear }.
-  4. Based on the form of the relationship as it is illustrated above, the _____ is quite { strong, weak }.

Reflect and discuss (B)

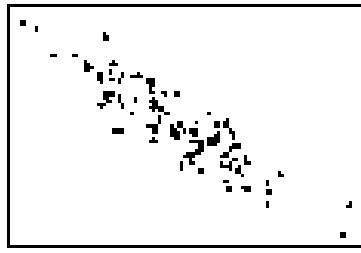
-  5. Which of the two scatterplots (see corresponding module reading) do you think displays a _____ linear relationship?
 - A. The top scatterplot.
 - B. The bottom scatterplot.
 - C. Both relationships are equally strong.

Reflect and discuss (C)

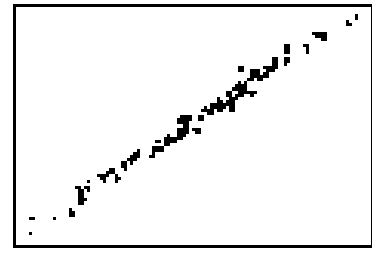
-  6. Match each scatterplot with the most likely _____ value listed below.



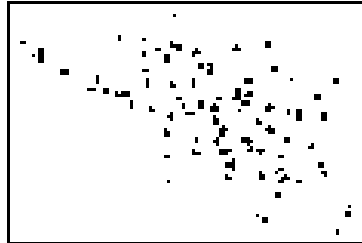
A.



B.



C.



D.

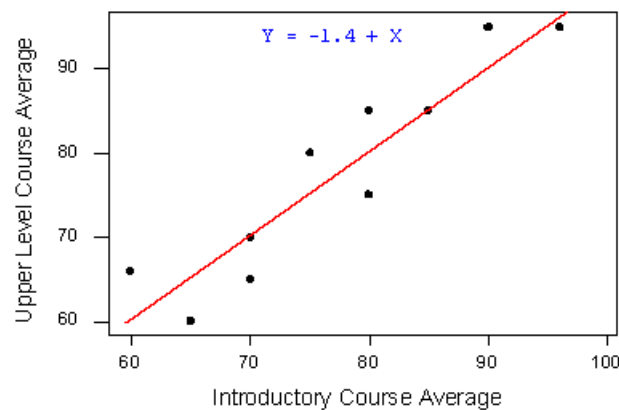


E.

- a. $r = 0.436$.
- b. $r = 0.100$.
- c. $r = -0.897$.
- d. $r = 0.995$.
- e. $r = -0.575$.

Reflect and discuss (D)

Background A statistics department is interested in tracking the progress of its students from entry until graduation. As part of the study, the department tabulates the performance of 10 students in an introductory course and in an upper-level course required for graduation. The scatterplot below includes the least squares line (the line that best explains the upper-level course average based on the lower-level course average), and its equation:



 7. What is the _____ of the regression line?

- A. 1.4.
- B. -1.4.
- C. 1.
- D. -1.

 8. Which of the following is the correct interpretation of the _____ of the regression line?

- A. A student who performs 1 point better than another on the upper-level course is likely to perform 1 point worse in the introductory course.
- B. A student who performs 1 point worse than another on the upper-level course is likely to perform 1 point better in the introductory course.
- C. A student who performs 1 point better than another in the introductory course is likely to perform $10 - 1.4 = 8.6$ points better in the upper-level course.
- D. A student who performs 1 point better than another on the introductory course is likely to perform 1 point better in the upper-level course.

 9. A student scored an average of 90 in the introductory course. What score can we expect for this student in the _____ course?

Reflect and discuss (E)

 10. Research has suggested that there is a correlation between Internet usage and obesity. Using this information, a fictional study found a correlation of $r = 0.79$ between the number of hours of daily Internet usage and Body Mass Index, a measure of body weight.

Which of the following is _____? (select all that apply.)

- A. There is a pretty strong positive linear relationship between the number of hours spent on the Internet and the Body Mass Index scores.
- B. Using the Internet causes people to gain weight.
- C. Physical activity is a possible “lurking” variable behind this relationship, since people who spend a large amount of time using the Internet may be more likely to be less physically active.

Reflect and discuss (F)

 11. The tourism industry hires a marketing company to promote overseas vacations. The marketing company’s research shows a $+0.72$ correlation between the number of overseas vacations booked and amount of free time, in days, people have each year. Therefore, the marketing company decides to develop an advertising campaign aimed specifically at people who have lots of free time.

Which of the following is _____ about this decision?

- A. The decision may not be a good one, since it is possible that there is a “lurking” variable, such as income, behind both the number of booked overseas vacations and the number of free days people have each year.
- B. The decision is a good one, since we know that the number of free days people have cause them to book more vacations.
- C. The decision is a good one, since there is a strong positive relationship between number of booked vacations and the number of free days people have each year.

Reflect and discuss (G) Causation

Background An Associated Press article captured the attention of readers with the headline “Night lights bad for kids?” The article was based on a 1999 study at the University of Pennsylvania and Children’s Hospital of Philadelphia, in which parents were surveyed about the lighting conditions under which their children slept between birth and age 2 (lamp, night light, or no light) and whether or not their children developed nearsightedness (myopia). The purpose of the study was to explore the effect of a young child’s nighttime exposure to light on later nearsightedness.

Here is the two-way table that summarizes the collected data:

Myopia Bed Light			
	Nearsighted	Not Nearsighted	Total
Lamp	41	34	75
Night Light	79	153	232
No Light	17	155	172
Total	137	342	479

The conditional percentages allow us to compare the distribution of later nearsightedness among children who were exposed to each of the three nighttime light levels:

Myopia Bed Light			
	Nearsighted	Not Nearsighted	Total
Lamp	54.7%	45.3%	100.0%
Night Light	34.1%	66.0%	100.0%
No Light	9.9%	90.1%	100.0%

The striking finding was that children who slept with lamps on were *more than 5 times more likely* to be nearsighted later in life (54.7% vs. 9.9%). Based upon this data alone, parents might discontinue using night-lights and lamps with young children.

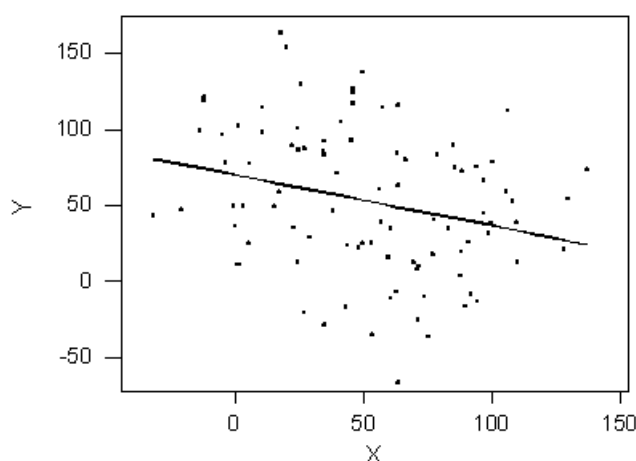
12. Do the data _____ evidence that early light exposure causes myopia?

13. What is a likely _____ in this case that could explain this observed relationship? (Hint: This lurking variable affects both the type of light with which a child sleeps and the child's later nearsightedness.)

Reflect and discuss (H) Causation

Recall that an instance of Simpson's paradox occurs whenever including a lurking variable causes you to rethink the direction of an association. In this activity, we will reinforce our understanding of Simpson's paradox.

The following scatterplot displays the relationship between two quantitative variables, X and Y :

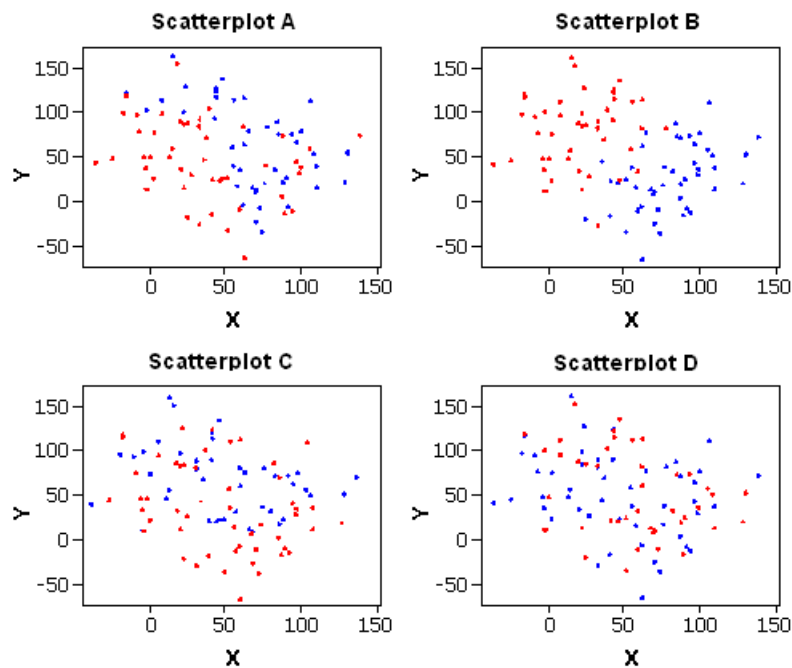


This graphical display indicates that the overall relationship between X and Y is *negative*.

In each of the four labeled scatterplots below, we included a different lurking variable, which separates the data points into two groups, blue points and red points. Your task is to look at all four displays and decide in which

case including the lurking variable leads to an instance of Simpson's paradox.

14. Which scatterplot is an example of _____?



- A. Scatterplot A.
- B. Scatterplot B.
- C. Scatterplot C.
- D. Scatterplot D.

Reflect and discuss (I)

15. A study was done on the graduation rates (graduate in 4 years versus did not graduate) from two universities: Mode College and Median University. Data were collected for the year 2012 in a variety of majors, and it was found that Mode College does better overall (i.e., has a higher percentage of students who graduate in 4 years). However, in each of the majors separately, Median University does better.

Which of the following is _____? (select all that apply)

- A. This is an example of Simpson's paradox.
- B. This is an example of a negative association between variables.
- C. This situation is mathematically impossible.
- D. "Type of major" could be a lurking variable in this example.